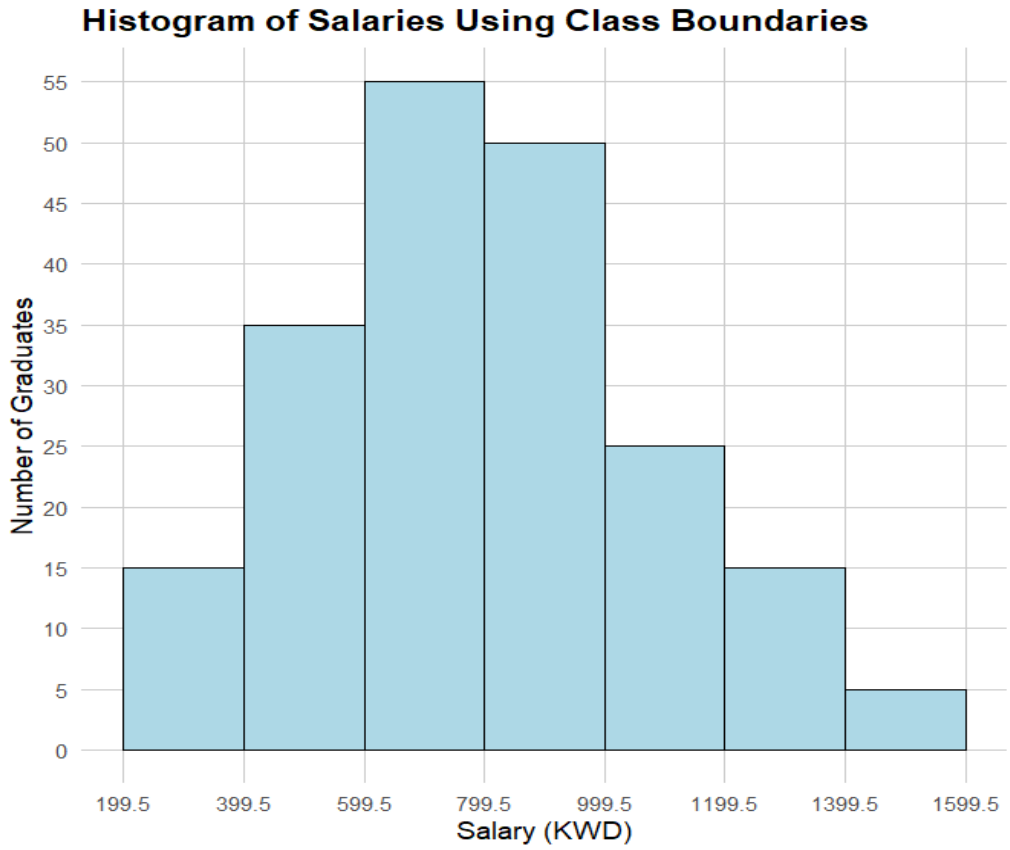
**TASK(A-1) BCS-4A**

**Q1:** **[Marks:1+(1+11)+2=15]**

A university collected the annual salaries from a sample of 200 graduates working as Data Scientists in different tech companies. The histogram below shows the distribution of their salaries (in KWD), grouped into bins. (200–399, 400–599, 600–799, 800–999, 1000–1199, 1200–1399, 1400-1599). The following tasks are given:

1. Estimate the percentage of graduates earning below 800 KWD.
   - ➢

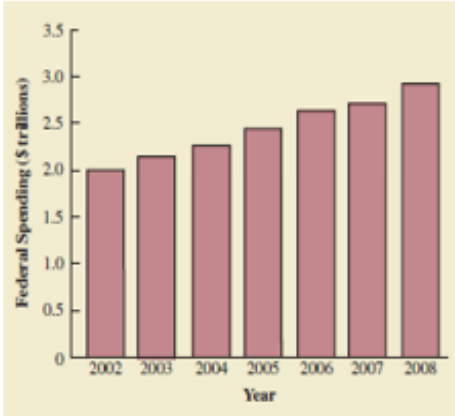
2. As a Data Scientist, explain why the exact skewness measure might be preferred by researchers instead of only relying on visualizations. Calculate and Comment on the exact skewness value of the salary distribution. [Given the Median is 781.32]




Below figure provides a bar chart showing the amount of federal spending for the years 2002 to 2008 (*USA Today*, February 5, 2008).

a) What is the variable of interest?

b) Are the data categorical or quantitative?

c) Are the data time series or cross-sectional?

d) Comment on the trend in federal spending over time.